

RATINGS OF PERCEIVED EXERTION DURING ACUTE RESISTANCE EXERCISE PERFORMED AT IMPOSED AND SELF-SELECTED LOADS IN RECREATIONALLY TRAINED WOMEN

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ABSTRACT

Cotter, JA, Garver, MJ, Dinyer, TK, Fairman, CM, and Focht, BC. Ratings of perceived exertion during acute resistance exercise performed at imposed and self-selected loads in recreationally trained women. *J Strength Cond Res* 31(8): 2313–2318, 2017—Resistance exercise (RE) is commonly used to elicit skeletal muscle adaptation. Relative intensity of a training load links closely with the outcomes of regular RE. This study examined the rating of perceived exertion (RPE) responses to acute bouts of RE using imposed (40% and 70% of 1 repetition maximum [1RM]) and self-selected (SS) loads in recreationally trained women. Twenty physically active women (23.15 ± 2.92 years), who reported regular RE training of at least 3 weekly sessions for the past year, volunteered to participate. During the initial visit, participants completed 1RM testing on 4 exercises in the following order: leg extension, chest press, leg curl, and lat pull-down. On subsequent visits, the same exercises were completed at the SS or imposed loads. The RPE was assessed after the completion of each set of exercises during the 3 RE conditions using the Borg-15 category scale. Self-selected loads corresponded to an average of approximately 57%1RM (± 7.62). Overall, RPE increased with load (40%1RM = 11.26 ± 1.95 ; SS 57% 1RM = 13.94 ± 1.58 ; and, 70%1RM = 15.52 ± 2.05). Reflecting the linear pattern found between load and perceived effort, the present data provide evidence that RPE levels less than 15 likely equate to loads which are not consistent with contemporary American College of Sports Medicine (ACSM) guidelines for enhancing musculoskeletal health which includes strength and hypertrophy. Women desiring increases

in strength and lean mass likely need to train at an exertion level at or surpassing a rating of 15 on the Borg-15 category. This article examined the modification of training load on perceived exertion, but other variables, such as the number of repetitions completed, may also be targeted to achieve a desired RPE. The primary understanding is that women who engage in RE may not self-select loads that are consistent with the ACSM recommendations for musculoskeletal health.

KEY WORDS female, acute exercise, RPE, training load, strength training, exercise adaptations

INTRODUCTION

Resistance exercise (RE) results in various musculoskeletal adaptations that range from satellite cell activation to increases in muscle oxygenation, endurance, strength, and size. These changes collectively demonstrate the remarkable plasticity of skeletal muscle. Multiple training variables, including intensity, rest intervals, and volume, contribute to the variance in type and magnitude of neuromuscular adaptation from RE (7,10).

Relative intensity, conceptualized in one manner as the percentage of a 1 repetition maximum (1RM), links closely with the outcomes experienced, and has been reported to account for 18–35% of the variance in the hypertrophic response to RE performed with higher training loads (6). It should be noted, however, that evidence promoting muscular changes resulting from training to failure using lower loads is appearing in the literature (14,15,17). In either case, quantifying intensity of RE using 1RM (measurement of muscle force expressed in relation to a maximum level) can be demanding, and potentially inappropriate in the case of isolation exercises. Moreover, novice trainees may not have the ability to produce an accurate 1RM, which could lead to inadequate prescription of training intensity (19). With the sizable influence of intensity on training outcomes, a simplistic assessment of intensity of an RE workout is critical.

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The Borg rating of perceived exertion (RPE) scale was developed to act as a simple tool to assess exercise intensity (1). Although the scale is widely used, it has primarily been implemented with aerobic exercise. It is important to recognize that RPE can be assessed at multiple, distinct points during an RE routine. For example, RPE could be assessed after the final repetition of a set, after the completion of all sets of an exercise, or at the conclusion of a workout session (2,16). Furthermore, RPE could be reported for the active muscle group or be used to summarize the intensity of the full routine by reporting an overall RPE. Considering the paucity of RE-RPE research, questions and knowledge of the utility and usage of the RPE scale during RE remain relatively limited.

Initial work by Gearhart et al. (9) established scaling procedures for RPE assessment during RE and found that fewer repetitions using heavier weight resulted in higher perceived effort compared with more repetitions using lighter weights (8). Employing women subjects, Lagally et al. (11) found that active muscle RPE and overall body RPE increased as the intensity of exercise increased (30%, 60%, and 90% of 1RM) when work was standardized. The authors also aligned the collection of RPE with blood lactate and electromyography concluding that RPE may be a useful technique to gauge and regulate the intensity of RE (12). In a randomized, crossover design, RPE was assessed in 20 subjects immediately after each set of 5 different exercises and 30 minutes postexercise (session RPE) at low, moderate, or high intensity. Corroborating previous evidence, as intensity increased, the RPE significantly increased (2).

These data offer evidence of the utility of RPE to monitor acute RE, but the extent to which RPE may differ between bouts characterized by loads that are self-selected (SS) or imposed during the training session is not established. Resistance exercise intensities that elicit perceptions of exertion that deviate considerably from levels of exertion that individuals would SS may detract from the efficacy of the training stimulus and motivation for subsequent RE participation.

We have previously demonstrated that untrained women selected to train with modest loads (56%1RM) that elicited lower levels of perceived exertion relative to imposed loads consistent with conventional RE prescription guidelines often inclusive of 1–3 sets of exercise ranging from 6 to 12 repetitions of an exercise (4). More recently, we demonstrated that although acute bouts of RE performed at SS and imposed loads yielded similar improvements in affective responses, trained women reported greater self-efficacy and intention, both established motivational correlates of exercise behavior, to participate in future RE using the SS load (5). There is need to investigate the RPE responses to SS and imposed loads among recreationally trained women using conventional prescriptions.

From a practical perspective, examining the RPE responses to SS and imposed load bouts of RE could help

inform fitness professionals in developing personalized RE programs that are tailored to individual preferences. Consequently, the purpose of this study was to examine the RPE responses to acute bouts of RE using imposed (40% and 70% of 1RM) and SS loads in recreationally trained women.

METHODS

Experimental Approach to the Problem

The focus of this work was to evaluate differences in RPE between imposed and SS load RE conditions. This study involved an ancillary analysis of RPE responses from a previous investigation of the affective response to acute RE (5). Although a detailed methodology of this study has been presented previously (5), a brief summary of the methods is provided here. During the initial visit, participants completed 1RM testing on 4 exercises in the following order: leg extension, chest press, leg curl, and lat pull-down. In 3 subsequent visits, the same exercises were completed at the SS or imposed loads. As described previously (4), subjects were provided with instructions on how to gauge their overall sense of effort level utilizing RPE.

Subjects

Twenty women primarily comprised of college age (M age = 23.15 ± 2.92 yrs.) and Caucasian race (90% of the sample) volunteered to participate. Each participant was physically active and reported regular RE training of at least three weekly sessions for the past year. Prior to participation in the study, each volunteer read and signed an informed consent document that had been previously approved by the Human Subjects Institutional Review Board at the Ohio State University.

Procedures

All exercises were performed using Cybex resistance training machines (Medway, MA, USA). One-repetition maximum testing was conducted using previously validated procedures (18) and was used to establish the imposed loads (40% and 70% of 1RM) and calculate the %1RM load selected in the SS RE condition. The order in which the 3 conditions were completed was randomly assigned, presented in a counter-balanced fashion, and completed on separate days at least 72 hours apart. The time of day the RE conditions were performed was standardized within, but not between, participants. Each RE condition comprised 3 sets of 10 repetitions of each exercise and a 2-minute recovery interval was maintained throughout the RE conditions.

The characteristics of the imposed loads were intentionally divergent when juxtaposed with contemporary resistance training guidelines. The intensity during one condition was typical of the load for enhancing muscular strength and hypertrophy (70%1RM), whereas the other condition was imposed well below that level (40%1RM). During the SS condition, the participants were instructed to select a load that would be comfortable, yet still provide a good, challenging workout. This method is consistent with procedures

TABLE 1. Descriptive statistics of the 1RM load and rating of perceived exertion responses after each set during the imposed and self-selected load resistance conditions.*

Time point	1RM load	40%1RM, M $\pm$ SD	70%1RM, M $\pm$ SD	Self-selected, M $\pm$ SD	COND effect, <i>p</i>	Set effect, <i>p</i>	INT effect, <i>p</i>
Leg Ext	174.50 $\pm$ 37.02				<0.001†‡	<0.001§ ¶	0.241
Set 1		11.9 $\pm$ 1.7	15.5 $\pm$ 1.8	12.8 $\pm$ 3.4			
Set 2		12.5 $\pm$ 1.8	16.0 $\pm$ 1.8	14.3 $\pm$ 1.1			
Set 3		12.8 $\pm$ 2.1	16.4 $\pm$ 2.0	14.7 $\pm$ 1.4			
Chest press	115.50 $\pm$ 20.96				<0.001†#	<0.001§ ¶	0.029**††
Set 1		10.0 $\pm$ 2.1	13.9 $\pm$ 1.8	13.1 $\pm$ 1.6			
Set 2		10.3 $\pm$ 2.1	14.7 $\pm$ 1.8	13.3 $\pm$ 1.5			
Set 3		10.6 $\pm$ 2.1	15.7 $\pm$ 2.0**††	14.2 $\pm$ 1.7**††			
Leg curl	127.00 $\pm$ 26.03				<0.001†‡#	0.001§¶	0.834
Set 1		12.2 $\pm$ 2.0	15.7 $\pm$ 2.2	14.0 $\pm$ 1.5			
Set 2		12.9 $\pm$ 2.2	16.6 $\pm$ 2.2	14.7 $\pm$ 2.2			
Set 3		13.2 $\pm$ 2.6	17.0 $\pm$ 2.3	15.0 $\pm$ 2.1			
Lat PD	91.00 $\pm$ 18.47				<0.001†‡#	<0.001§¶	<0.001**††
Set 1		9.4 $\pm$ 2.0	14.0 $\pm$ 2.0	12.8 $\pm$ 1.4			
Set 2		9.7 $\pm$ 2.1	15.0 $\pm$ 2.2**	13.8 $\pm$ 1.6**			
Set 3		9.7 $\pm$ 2.0	15.9 $\pm$ 2.5**††	13.9 $\pm$ 1.8**			
Average		11.26 $\pm$ 1.95	15.52 $\pm$ 2.05	13.94 $\pm$ 1.58			

*1RM = 1 repetition maximum; COND = condition; INT = interaction; EXT = extension; PD = pull-down.
†Significance at $p \leq 0.05$ is 40%1RM vs. 70%1RM.
‡Significance at $p \leq 0.05$ is 70%1RM vs. SS.
§Significance at $p \leq 0.05$ is Set 1 vs. Set 2.
||Significance at $p \leq 0.05$ is Set 2 vs. Set 3.
¶Significance at $p \leq 0.05$ is Set 1 vs. Set 3.
#Significance at $p \leq 0.05$ is 40%1RM vs. SS.
**Significance at $p \leq 0.05$ is different from Set 1.
††Significance at $p \leq 0.05$ is different from Set 2.

used in previous RE studies incorporating SS loads (3,4). Participants were allowed to choose and adjust the load for each set lifted during the SS condition. RPE was assessed following the completion of each set of exercises during each of the three RE conditions using the Borg-15 category scale (1).

Statistical Analyses

The RPE responses accompanying each of the 4 exercises were analyzed by separate 3 (Condition: 40%1RM; 70% 1RM; SS) $\times$ 3 (Set: first, second, third) repeated-measures analyses of variance. If the assumptions of sphericity were violated (checked with Mauchly's W), the Huynh-Feldt adjusted statistic was used. The Sidak adjustment was used in post hoc analysis to determine the presence of significant mean differences. Effect sizes (Cohen's d) were also calculated by taking the mean difference and dividing by the pooled SD .

RESULTS

Women in the study selected loads that corresponded to an average of approximately 57%1RM (± 7.62) during the SS load condition. The average RPE during the conditions are shown in Table 1 and were as follows: 40%1RM = 11.26 (± 1.95), 70%1RM = 15.52 (± 2.05), and SS = 13.94 (± 1.58).

Leg Extension

Analysis of the RPE responses to the leg extension yielded significant main effects for Condition ($p < 0.001$) and Set ($p < 0.001$), but not the interaction ($p = 0.241$). Post hoc analysis of the Condition main effect revealed that RPE was significantly higher in the 70%1RM vs. 40%1RM ($M = 15.92$ vs. 12.40 , $d = 1.87$) and 70%1RM vs. SS ($M = 15.92$ vs. 13.92 , $d = 0.94$) Conditions. Post hoc analysis of the Set main effect demonstrated that RPE significantly increased from Set 1 to Set 2 ($M = 13.38$ vs. 14.23 , $d = 0.34$), Set 2 to Set 3 ($M = 14.23$ vs. 14.62 , $d = 0.17$), and Set 1 to Set 3 ($M = 13.38$ vs. 14.62 , $d = 0.47$). The descriptive statistics for the RPE responses to the leg extension are provided in Table 1.

Chest Press

Significant main effects for Condition ($p < 0.001$), Set ($p < 0.001$), and Condition $\times$ Set interaction ($p = 0.029$) were found for the RPE responses on the chest press. Post hoc analysis of the Condition $\times$ Set interaction revealed that RPE increased from Set 1 to 3 and Set 2 to 3 in the 70% 1RM (Set 1 to Set 3, $M = 13.90$ vs. 15.65 , $d = 0.92$; Set 2 to Set 3, $M = 14.65$ vs. 15.65 , $d = 0.52$) and SS Conditions (Set 1 to Set 3, $M = 13.10$ vs. 14.15 , $d = 0.63$; Set 2 to Set 3, $M = 13.30$ vs. 14.15 , $d = 0.53$), but there was no change during the 40%1RM Condition. The RPE was significantly higher in the 70%1RM vs. 40%1RM ($M = 14.73$ vs. 10.28 , $d = 2.18$) and SS vs. 40%1RM ($M = 13.52$ vs. 10.28 , $d = 1.72$) Conditions. The descriptive statistics for the RPE responses to the chest press are provided in Table 1.

Leg Curl

The RPE responses to the leg curl yielded significant main effects for Condition ($p < 0.001$) and Set ($p = 0.001$), but not the interaction ($p = 0.834$). The RPE was significantly higher in the 70%1RM vs. 40%1RM ($M = 16.40$ vs. 12.73 , $d = 1.63$), 70%1RM vs. SS ($M = 16.40$ vs. 14.52 , $d = 0.41$), and SS vs. 40%1RM ($M = 14.52$ vs. 12.73 , $d = 0.84$) Conditions. Additionally, post hoc analysis of the Set main effect demonstrated that RPE significantly increased from Set 1 to Set 2 ($M = 13.93$ vs. 14.70 , $d = 0.31$) and Set 1 to Set 3 ($M = 13.93$ vs. 15.02 , $d = 0.42$). The descriptive statistics for the RPE responses to the leg curl are provided in Table 1.

Lat Pull-Down

Analysis of variance of the RPE responses to the lat pull-down yielded significant main effects for Condition ($p < 0.001$), Set ($p < 0.001$), and Condition $\times$ Set interaction ($p < 0.001$). Post hoc analysis of the Condition $\times$ Set interaction revealed that RPE increased across all sets in the 70% 1RM Condition (Set 1 to Set 2, $M = 13.95$ vs. 14.90 , $d = 0.49$; Set 2 to Set 3, $M = 14.90$ vs. 15.65 , $d = 0.36$; Set 1 to Set 3, $M = 13.95$ vs. 15.65 , $d = 0.83$) select sets during the SS Condition (Set 1 to Set 2, $M = 12.80$ vs. 13.80 , $d = 0.66$; Set 1 to Set 3, $M = 12.80$ vs. 13.90 , $d = 0.68$), but did not increase in the 40%1RM Condition. When collapsed across sets, RPE was significantly higher in the 70%1RM vs. 40%1RM ($M = 14.93$ vs. 9.58 , $d = 2.44$), 70%1RM vs. SS ($M = 14.93$ vs. 13.50 , $d = 0.70$), and SS vs. 40%1RM ($M = 13.50$ vs. 9.58 , $d = 2.13$) Conditions. The descriptive statistics for the RPE responses to the lat pull-down are provided in Table 1.

DISCUSSION

The purpose of this study was to examine the RPE responses to acute bouts of RE using imposed (40% and 70% of 1RM) and SS loads in recreationally trained women. We have previously demonstrated that affective and motivational outcomes to acute RE are load dependent (5). The primary finding of our ancillary RPE analysis revealed that, when permitted to choose the load in the SS condition, participants selected a load of approximately 57% of 1RM that was accompanied by an average RPE of 13–14 ("Somewhat Hard"). Conversely, in the imposed 70%1RM condition, average RPE ratings of 15–16 ("Hard") were observed. The linear pattern of increasing RPE and load across the conditions is consistent with the findings of previous studies (16,19). Thus, these findings provide further evidence supporting the accuracy and utility of RPE during acute RE. Monitoring RE intensity using the Borg-15 category scale seems to be a feasible method.

Common imposed loads may not align with SS loads in this group of RE participants. Recreationally trained women chose a lighter load and reported lower average RPE than are typically prescribed to elicit improvements in muscular strength and hypertrophy (i.e., 70%1RM). Collectively, these findings suggest that loads imposed in traditional RE

training prescriptions may not be representative of the load and exertion levels women choose to train at during their own self-directed RE sessions. The present findings are consistent with those of our previous research in untrained women (4) and extend these results to recreationally trained women.

Further investigation of the data by exercise demonstrates significant interaction effects with upper body, but not lower body, exercises. This was primarily driven by increases in RPE across sets in 70% and SS conditions with an absence in the 40% condition. Several physiological factors may contribute to these observed differences, such as muscle mass involvement, neuromuscular control, or blood flow. Additionally, it is plausible that psychosocial variables such as perceived challenge or preference for upper vs. lower body exercises may have influenced the RPE responses. However, the factors driving the observed interaction effect cannot be determined from data collected in this study warranting further inquiry.

Interpretation and application of results should be examined in context to the following limitations. The number of repetitions to be completed in each set was fixed at 10. The authors chose to limit repetitions in such a way based on the American College of Sports Medicine (ACSM) recommending “an individual to complete 8–12 repetitions per set” (13). Although further recommendations suggest that “each set should be performed to the point of muscle fatigue,” it cannot be assumed that the general population purposefully interprets this recommendation. Indeed, when recreationally trained participants were able to SS their load, a load was chosen below what was needed to achieve a fatigued state. An additional limitation may be that participants did not have sufficient resistance training experience to meet the suggested element of fatigue. The authors chose to recruit recreationally trained women with at least 1 year of experience in resistance training 3 times a week, but it is currently unknown what level of experience is required to select adequate loads for this intended purpose. Additionally, during the SS sessions, participants were free to choose a different load during each of the 3 sets. Given that an inappropriate choice (either too light or too heavy) occurred in the first set, the opportunity to change the load at sets 2 and 3 was provided.

Moreover, the authors are not advocating that a goal of 10 repetitions is the sole way to achieve physiological adaptations to resistance training. This goal was chosen as it aligns with the current recommendations of the ACSM (13). In fact, recent research has demonstrated that similar hypertrophic gains and strength increases may result from both low- and high-load resistance training protocols (14,15,17). The recreationally trained women in this study generally failed to SS a load that elicited a fatigued state when an imposed repetition goal was set. It is unknown whether an imposed fatigue goal set at a given intensity would result in SS repetitions leading to termination of the set before a fatigued state occurred.

It is important to note that in our previous study focusing on the affective and motivational outcomes (5), we demonstrated that women reported their greatest intention and self-efficacy for future resistance training participation with the SS load. It is well established that self-efficacy and intention represent primary motivational correlates of exercise participation. Accordingly, the present ancillary analysis of the RPE responses from that study reveals that exertional ratings corresponding to “Somewhat Hard” represented the load and effort level that recreationally trained women cited as being most willing to use during future self-directed resistance training.

PRACTICAL APPLICATIONS

The practical application of the present findings is that recreationally trained women SS a load which results in an exertion level perceived to be “Somewhat Hard.” This load may differ from loads commonly implemented in RE prescriptions to increase muscular strength and hypertrophy. Consequently, these findings can be used to assist in personalizing RE prescriptions by implementing approaches that emphasize appropriate, goal-directed, load progression. Professionals should be aware that recreationally trained women may have concerns regarding the use of heavier loads and any exertion-related discomfort as represented by their SS loads.

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